

LECTURE 3 — COAL COUNTRY AND THE BATTLE OF BLAIR MOUNTAIN

PHY199-RAP · Energy in the 21st Century

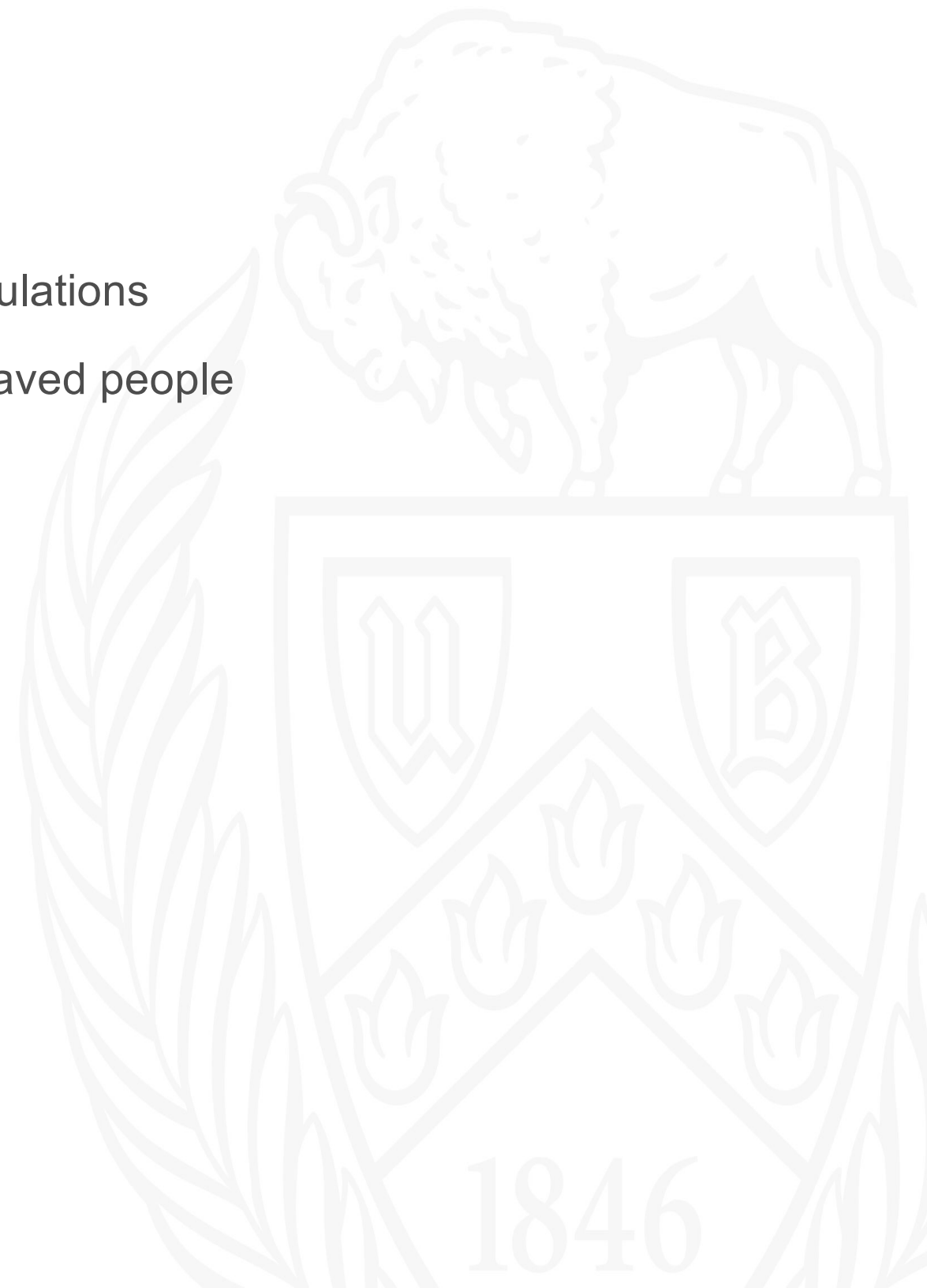
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 **University at Buffalo**® The State University of New York

1846

Content Warnings

- Genocide and colonialism against Native American populations
- Abuse and death of enslaved people and escaped enslaved people
- White supremacy organizations
- Child labor
- Exploitation and suppression of workers



THE RESOURCES



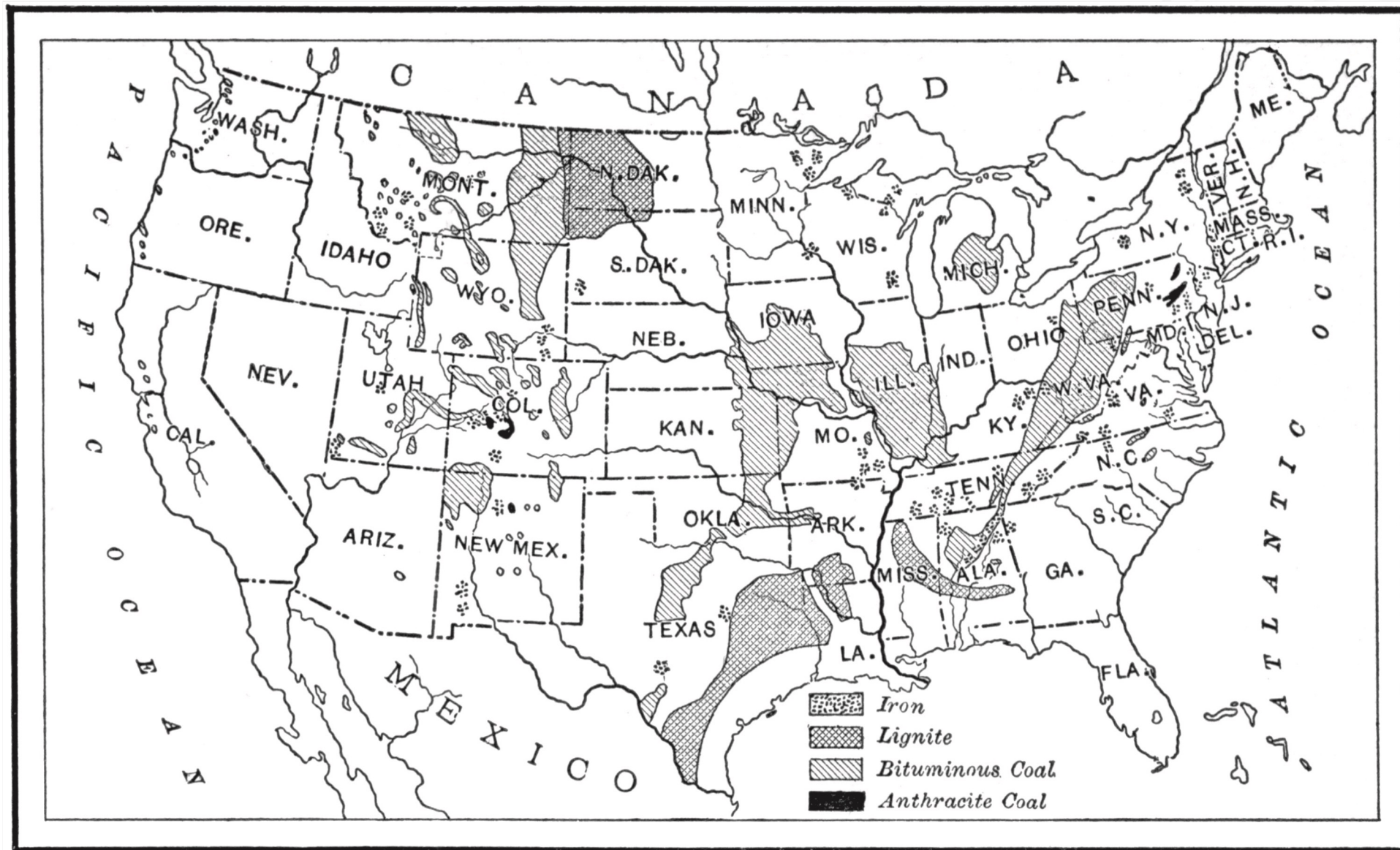
How to Make Steel

- Iron + coal = steel



<https://thatvideogameblog.com/blast-furnace-recipe/>

How to Make Steel



Coal and Iron Deposits in the United States, 1910
Ralph S. Tarr, B.S., F.G.S.A. and Frank M. McMurry, Ph.D., New Geographies 2nd ed
(New York, NY: The Macmillan Company, 1910)
Downloaded from *Maps ETC*, on the web at <http://etc.usf.edu/maps> [map #02085]

How to Make Steel



Coal

- Bituminous :
 - Very abundant
 - Used for production of steel and also for power generation
 - Burns with lots of outgassing and pollution
- Anthracite :
 - Very rare (<1% of coal)
 - Used for power generation and high-grade metallurgy
 - Mostly coming from area surrounding Lehigh Valley in PA



Iron Ore

- [Magnetite](#)
- [Hematite](#)
- Mostly came from Minnesota
- Almost all used to make steel

Magnetite



Hematite



Mining Coal and Iron

Mining in Minecraft



- Fun
- Has zombies
- Can fight Ender Dragon

Mining IRL



Breaker boys and men, Pennsylvania Coal Co. Photo Lewis Hine. PD (LoC).

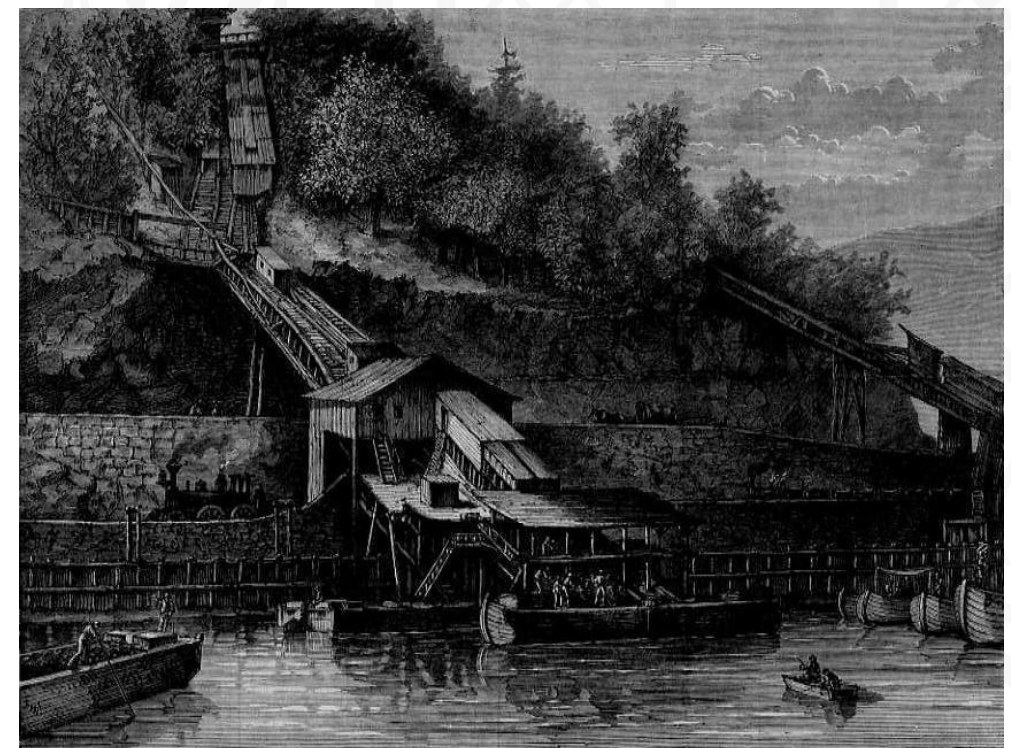
- Not fun
- Has coal lung
- Can fight your... boss?

Anthracite Coal

- Pennsylvania's [Coal Region](#) — Carbon, Lackawanna, Luzerne, Northumberland, Schuylkill counties — holds **the largest known anthracite deposits in the world**: roughly **7 billion short tons**.
- The [Lehigh Valley](#) in the Appalachian range is where that coal got out:
 - **Lehigh Canal**, opened **1827** — coal to Philadelphia and New York
 - **Lehigh Valley Railroad** (Asa Packer) — coal to everywhere else
 - **Bethlehem Steel**, founded **1899** — the nation's #2 steelmaker, built on that coal



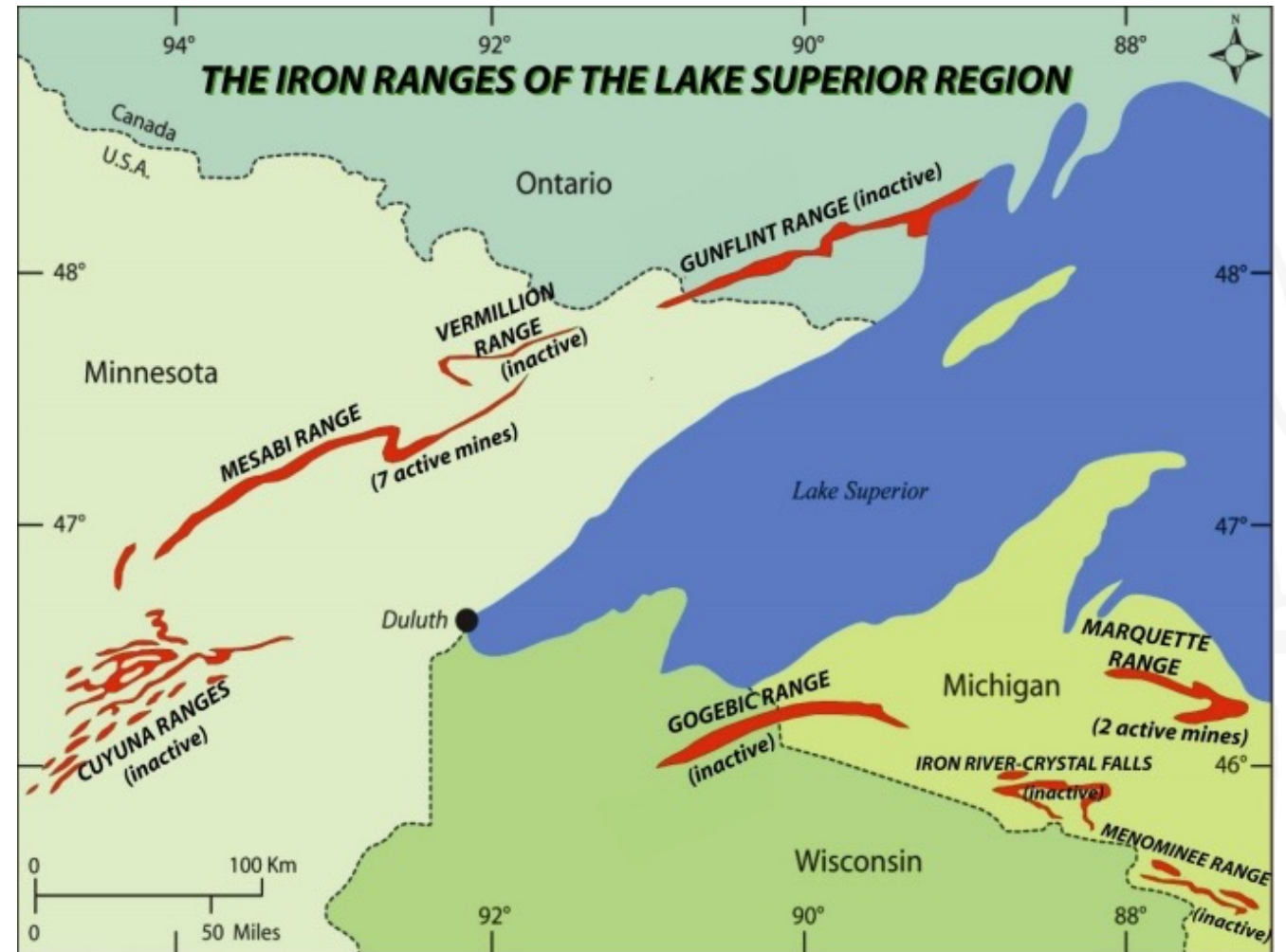
Breaker boys and men, Pennsylvania Coal Co. Photo Lewis Hine. PD (LoC).



Chutes loading coal boats on the Lehigh Canal (opened 1827). PD.

Iron Ore

- Primarily from the Mesabi Range in northeastern Minnesota
 - Ojibwe in N.E. Minnesota knew of the iron ore in the region, and told blacksmith N.A. Posey in 1863 and surveyor George Stunz
 - Immigrants from Europe (German, Irish, Swedes, Italians) moved into the region near Duluth in the late 19th Century
- Mining companies opened in the late 19th century, examples:
 - Soudan Mine
 - Mountain Iron Mine
- Carnegie, Morgan, Charles Schwab, Elbert Gary and others eventually merged all of the disparate steel operations into one megacorporation, US Steel, which held a virtual monopoly on steel production in the US



THE PEOPLE, THEIR UNIONS, AND THE COMPANY TOWN



Who dug it, and where the wealth went

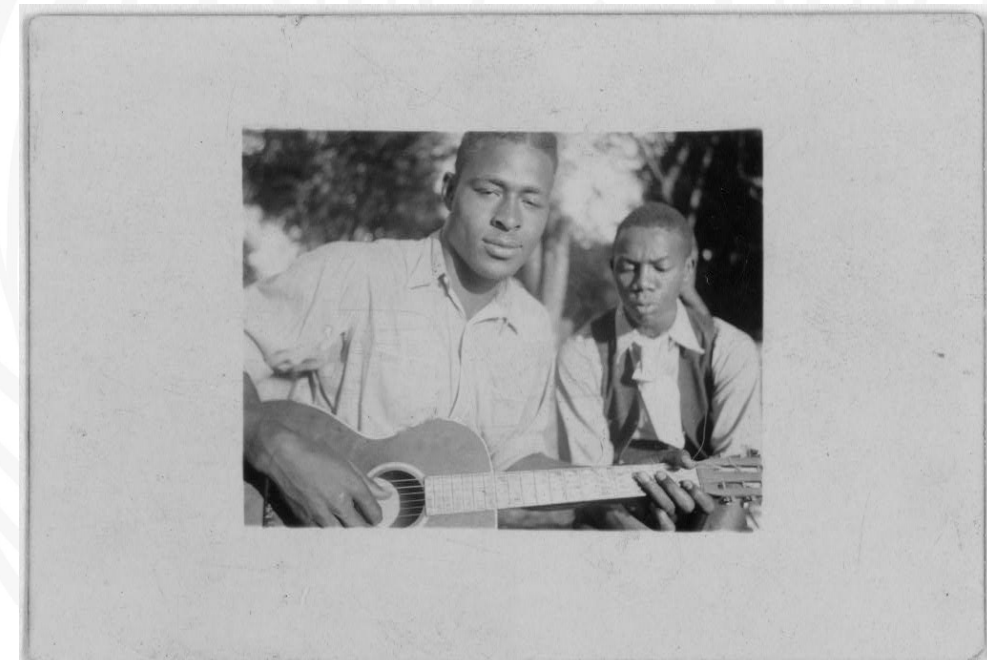
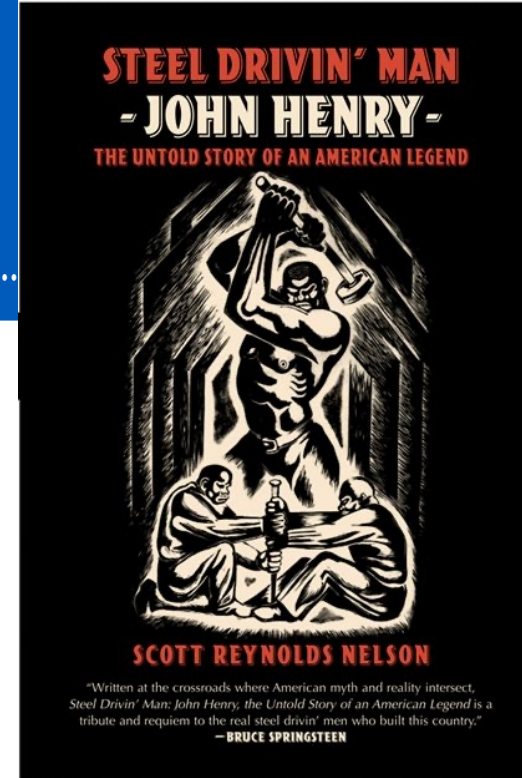
- The Coal Region's workforce was imported in waves: **English, Welsh, Irish, German** first, then **Polish, Slovak, Ukrainian, Hungarian, Italian, Russian, Lithuanian and Jewish** immigrants, as well as extensive migration of **Black Americans** escaping the Jim Crow South in the Great Migration
- The coal did not stay. It went to the **Steel Belt** — Pittsburgh, Cleveland, Detroit, Chicago, and **Buffalo** — where Appalachian coking coal met Minnesota iron ore over Great Lakes shipping.



“Breaker boys” in a Kingston, Pennsylvania, anthracite coal mine in 1900. Predominantly immigrants, these boys often worked fourteen- to sixteen-hour days separating slate from the coal after it was taken out of the mine shafts. (*Library of Congress*)

John Henry

- **John Henry** is a Black American folk hero. According to legend, he raced a steam drill through a mountain at Big Bend Tunnel, won, and died with the hammer in his hand.
- Historian **Scott Reynolds Nelson** traced him to prison records — a **19-year-old** named John William Henry, in the Virginia Penitentiary, sent to tunnel work on the **Chesapeake & Ohio Railway** in the 1870s.
 - **Convict laborer**, and he probably died of **silicosis** from rock dust — not a burst heart.
 - John Henry stands for hundreds who died anonymously.
 - **Convict leasing** put thousands of Black men, arrested on minor or invented charges, into mines and tunnels across America,
 - For-profit prisons and leasing prisoners is a practice that still exists today
- [The Ballad of John Henry](#) became a very [popular folk song](#), Sung by many including Gabriel Brown and Rochelle French, as well as 21st century artists like [Joe Bonamassa](#) (From Rochester)



Gabriel Brown and Rochelle French, Eatonville, Florida

Unions

- The brutal mining conditions, awful health outcomes, and exploitation of workers and the community led to the formation of the [United Mine Workers of America](#) in 1890
- Modeled after the [American Federation of Labor](#) (AFL), but then broke with AFL to set up its own federation, the [Congress of Industrial Organizations](#), later all incorporated into the [AFL-CIO](#)



<https://umwa.org>

Slave Catchers and Private Detective Agencies

- Before the Civil War, [slave catching](#) was a regular occupation in the US, especially the South
 - Recruited militia from places like [The Citadel](#) and the [Virginia Military Institute](#)
 - Became a paramilitary operation
 - After the Civil War, led / linked to the formation of the [Ku Klux Klan](#)
 - These were still implicitly and explicitly enforced by Southern police officers
 - Addressed in popular culture, such as [Sinners \(2025\)](#) where Smoke and Stack confront KKK members who come to kill them after selling them the juke joint
- Companies needed a private military force, hired existing agencies to infiltrate and break up unions
 - [Baldwin-Felts](#)
 - [Pinkertons](#)
- This led to a **ready-made force of paramilitary thugs** who were deputized and enabled by the Federal and local governments to brutally suppress the workforce, Black people escaping the Jim Crow south, and immigrants from everywhere



Depiction of Mississippi slave patrol



Michael B. Jordan as Smoke, killing a member of the KKK in the 2025 film "Sinners"



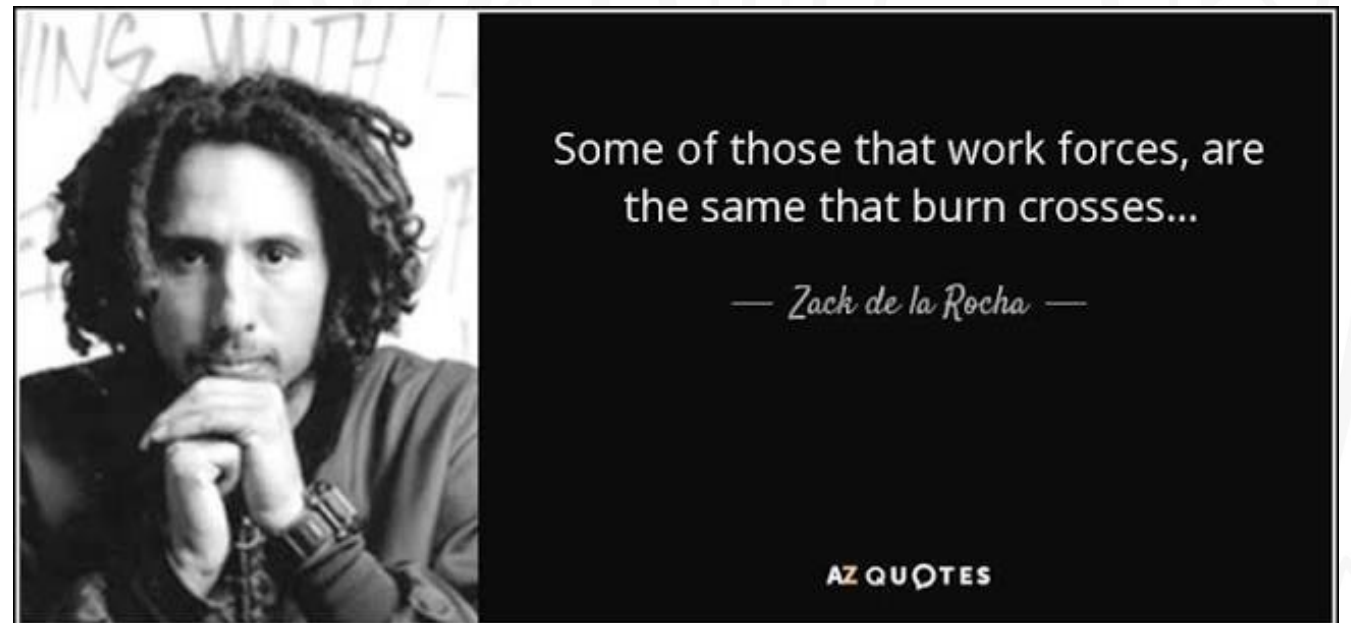
A Baldwin Felts agent at [Glen Jean](#), [Fayette County](#), 1902.

The Revival of the Ku Klux Klan

- In the same years, the **second Ku Klux Klan** was the fastest-growing membership organization in America.
 - Refounded **24 December 1915** at Stone Mountain, GA, inspired by the film *The Birth of a Nation*
 - Peak membership by **1924: an estimated 3-8 million**
 - **Not just the South.** Detroit had ~40,000 members. In Indiana, ~18% of eligible men joined
 - Targets: Black Americans, Catholics, the Jewish diaspora, and Southern and Eastern European immigrants
- During that period, many law enforcement members were also part of the KKK
 - Inspired the lyrics of the popular song “Killing in the Name Of” by Rage Against the Machine



Klan parade, Pennsylvania Avenue, Washington DC, 1928. PD (NARA).



Lackawanna Steel

- The [Seneca nation](#) sold part of the [Buffalo Creek Reservation](#) in 1842
 - Part of this became the town of Seneca
 - 1899-1902 [Lackawanna Steel](#) moved from Scranton, in the Coal Region, to the Lake Erie shore south of Buffalo and renamed part of Seneca and we got West Seneca and Lackawanna in the end
 - Consolidated into US Steel in 1907
- Stated reasons included shipping costs — **and the fact that the United Mine Workers of America had organized Scranton in 1897 and struck successfully in 1900 and 1902.**
- Acquired by [Bethlehem Steel](#) in 1922
- By the 1940s that plant employed **over 20,000 people**, and closed in 1982 after massive layoffs
- The company had considerable friction with the neighboring communities due to public health problems including cholera, typhoid, and flu, and incorporated Lackawanna, NY in 1909



Inside the Lackawanna plant, Buffalo. PD (HAER, LoC).

Panic of 1907

- The “Knickerbocker Crisis” of 1907 was a three-week-long financial crisis that caused a recession and the NYSE fell 50%
- Triggered by an attempt to corner the market of the United Copper Company (not steel, but also had its own mining issues and exploitation)
- JP Morgan stabilized the financial system and the action was approved by US President Theodore Roosevelt



[Wall Street](#) during the bank panic in October 1907. [Federal Hall National Memorial](#), with its statue of [George Washington](#), is seen on the right.

The Company Store

A **company town** is a settlement where one company owns the housing, the store, the school, the church, and the job.

At their peak: **more than 2,500** company towns in the US, housing **about 3%** of the population. A **1916** Bureau of Labor Statistics survey found **34.4%** of the surveyed workforce living in company housing.

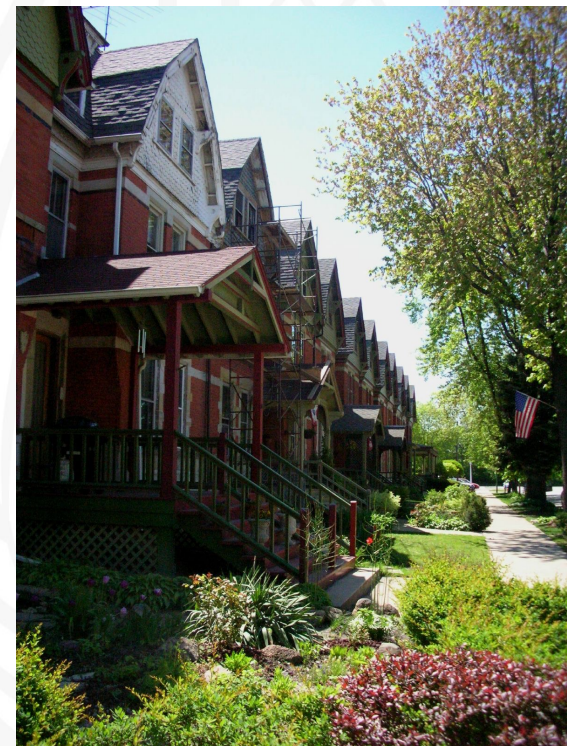
The company called this **paternalism** and considered it a moral duty. A federal commission investigating the **Pullman** strike of **1894** called it something else:

"The aesthetic features are admired by visitors, but have little money value to employees, especially when they lack bread."

Its verdict on Pullman's paternalism: **"Un-American."** Illinois sued, and in **1898** the state Supreme Court forced Pullman to sell the town.



The company store, Federal No. 1 Mine, Grant Town, WV. PD (NARA).



Pullman company housing, Chicago, today. Photo Richie Diesterheft, CC BY 2.0.

Scrip: private money

- Miners were often paid in **scrip** — basically 19th century crypto. Metal or paper tokens issued by the coal company.
 - Worth **full face value only at the company store**
 - Companies **refused to redeem it for cash**
 - Independent traders bought it at a discount of **25% and up**; nearby stores took it at **10-30% off**
 - So taking your pay somewhere else meant taking a pay cut of up to a quarter
- West Virginia banned scrip-as-wages in **1891** and made it non-transferable in **1925**. It survived until the coal industry itself collapsed in the **1950s**.
- **"I owe my soul to the company store"** is not a metaphor. It is a description of a currency system.



Coal scrip, Killarney, West Virginia. PD.

Sixteen Tons

- Song written by Merle Travis about the life of a coal miner in Rosewood, Kentucky
 - Travis grew up in Rosewood, and sung often about the exploitation of workers in the 19th and 20th centuries
- “Saint Peter don’t call me, cuz I can’t go... I owe my soul to the Company store”

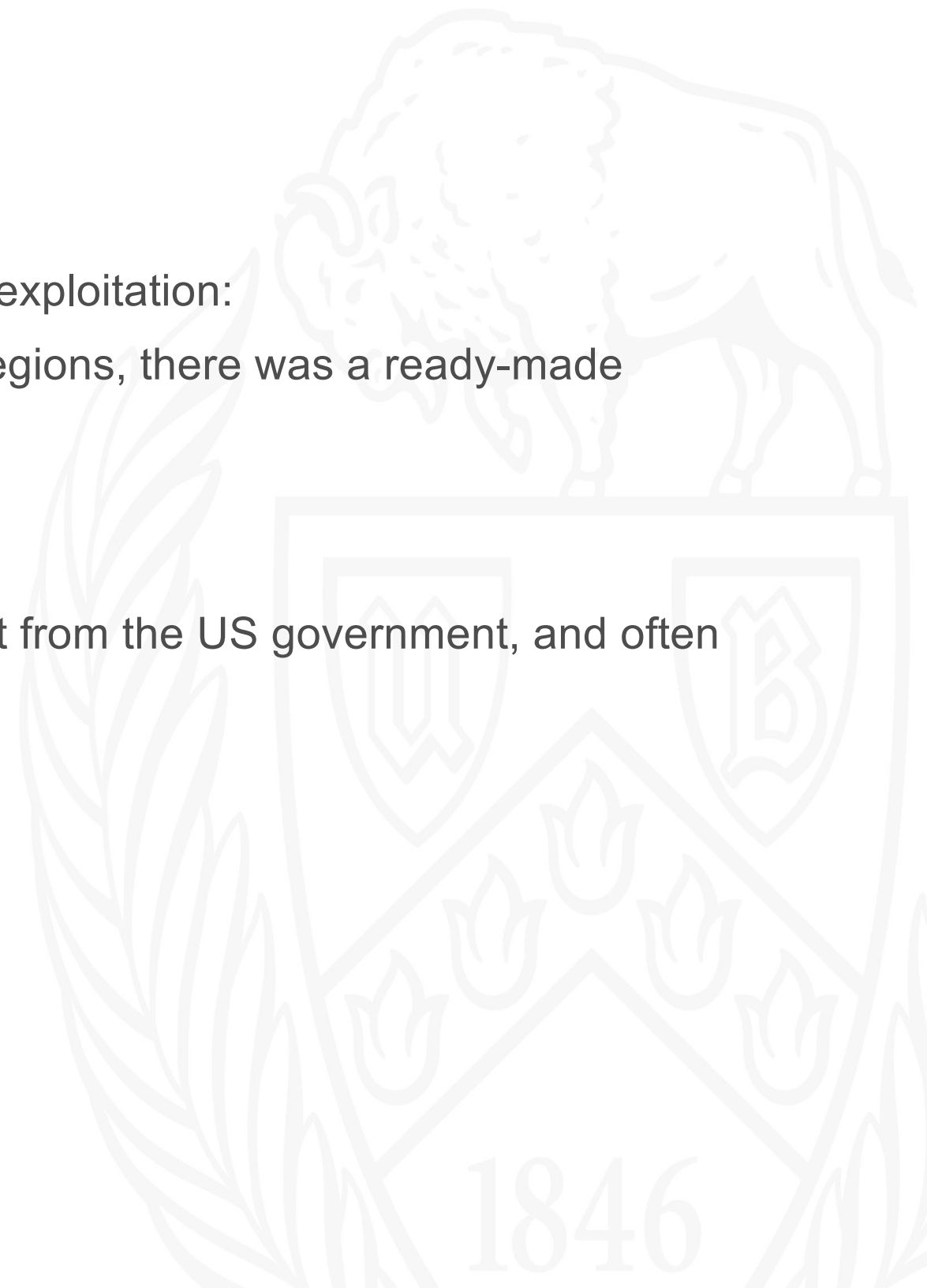


THE SHOOTING WAR



Conflicts Arise

- The conditions in the 19th and 20th century were rife for exploitation:
 - After expelling the Native American nations in the regions, there was a ready-made workforce of Black Americans and immigrants
 - Poor, desperate, and often unhoused
 - They took what they had to in order to live
 - Rapidly expanding industry with little to no oversight from the US government, and often direct and indirect support from them
- Classic case of the “haves” versus the “have nots”



1890-1930: the Coal Wars

Year	Event	What happened
1894	Cripple Creek strike (CO)	Miners vs. hired deputies
1902	Anthracite strike (PA)	150,000+ out for 163 days
1912-13	Paint Creek-Cabin Creek (WV)	Mine guards, an armored train, martial law
1914	Ludlow Massacre (CO)	~19-25 dead, including 12 children
1920	Matewan Massacre (WV)	10 dead in a 15-second gunfight
1921	Battle of Blair Mountain (WV)	~10,000 armed miners
1931-39	Harlan County War (KY)	"Bloody Harlan"



Ruins of the Ludlow tent colony, Colorado, April 1914. PD (LoC).

1890-1930 : The Coal Wars

- Not fought over wages, primarily. **The right to have a union at all.**
- To get there, miners had to get past:
 - **Yellow-dog contracts** — sign away your right to join a union as a condition of employment
 - **Mine guards** — **private security agents**, paid by the operators, **deputized by the county**
 - **Eviction** — you live in the company's house, so a strike makes you homeless the same week
 - **Scrip** — you are paid in the company's private money, worth face value *only at the company store*
- A miner in 1921: after loading car after car of coal, you still "couldn't come out to the company store and get enough food to feed your family."



Union leaders during the armed march, southern West Virginia, 1921. PD.

19 May 1920 → 1 Aug 1921: the fuse

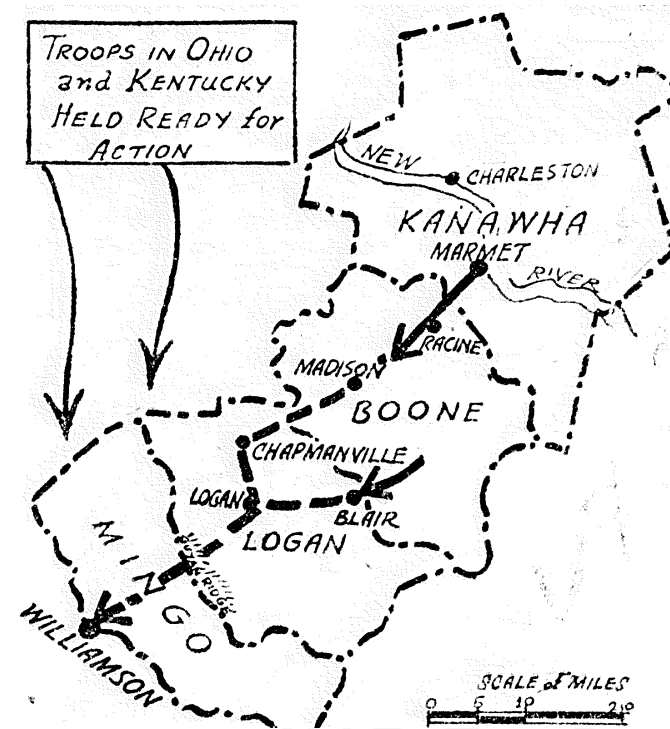
- **Matewan, WV, 19 May 1920.** Baldwin-Felts agents come to evict miners' families. Police chief **Sid Hatfield** and Mayor Cabell Testerman confront them. Fifteen seconds of shooting: **ten dead**, including the mayor and Albert Felts.
- Hatfield becomes a folk hero. Tried for murder in January 1921 — **and acquitted**.
- **1 August 1921.** Hatfield walks up the courthouse steps in Welch, unarmed, with his wife on his arm. Baldwin-Felts agents shoot him and his friend Ed Chambers dead on the steps.
- **Nobody is ever convicted.**
- Two weeks later, ten thousand miners pick up rifles.



Sid Hatfield, chief of police of Matewan. PD.

25 Aug - 2 Sep 1921: Blair Mountain

- ~10,000 miners marched on Logan County. Against them: ~3,000 "Logan Defenders" under Sheriff **Don Chafin**, paid by the **Logan County Coal Operators Association**, plus state police.
- Five days of fighting along a mountain ridge. Roughly **one million rounds** fired. Machine guns on both sides.
- Private planes, hired by Chafin, dropped homemade bombs on American citizens.** Army bombers flew surveillance under Gen. Billy Mitchell.
- It ended on **2 September** when federal troops arrived — **because the miners would not shoot at US soldiers.** Many had been US soldiers three years earlier.
- Casualty figures are contested:** Wikipedia gives 50-100 miners killed; the WV Mine Wars Museum documents ~16 confirmed.



The miners' march on Logan County. PD.



A machine-gun nest at Blair Mountain. PD.



A miner of the armed march. PD.



The Washington Times, 1921. PD.

Big Coal Won the War

- Let us be completely clear about the outcome.
 - **985 miners indicted** — murder, conspiracy, and treason against the **State of West Virginia**
 - **UMWA membership in the region collapsed from 50,000+ to about 10,000**
 - **80% of mines reopened** with replacement workers and yellow-dog contracts
 - **Union organizing in southern West Virginia was dead until 1933**
 - **Nobody was ever punished for killing Sid Hatfield.**
Nobody was punished for Ludlow either.
- The operators kept the coal, kept the towns, kept the money, and kept the courts.
- **The people who committed the atrocities got richer.**
That is the actual ending.



Miners display a bomb dropped on them from the air. PD.



Bill Blizzard - tried for treason, acquitted. PD.

POPULAR CULTURE AT THE TIME



The Muckrakers

The same decades produced journalists who made the abuses impossible to ignore — **muckrakers**, a name Roosevelt gave them as an *insult* on **14 April 1906**, borrowed from *Pilgrim's Progress*.

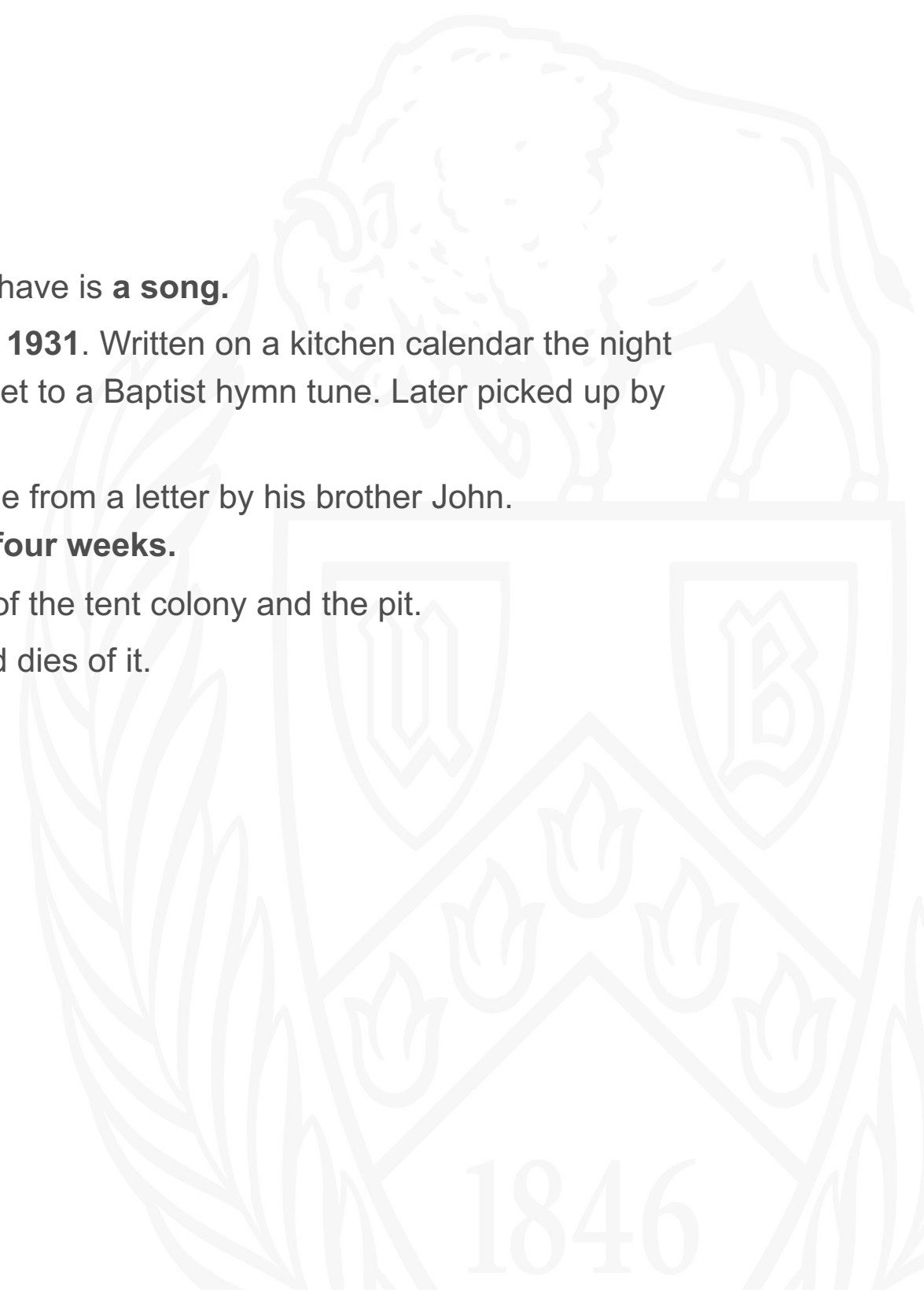
- **Ida B. Wells** — *Southern Horrors* (1892), *The Red Record* (1895) → documented lynching
- **Ida Tarbell** — *The History of the Standard Oil Company* (1902-08) → fed the 1911 breakup
- **Upton Sinclair** — *The Jungle* (1906) → Pure Food and Drug Act; Meat Inspection Act
- **Jacob Riis** — *How the Other Half Lives* (1890) → tenement reform
- **Lincoln Steffens** — *The Shame of the Cities* (1904) → municipal corruption prosecutions
- **Ray Stannard Baker** — *"The Right to Work"* (1903) → wrote specifically on **coal mining**



Ida B. Wells (1862-1931). PD.

Songs as primary sources

- When you own the press, the courts and the police, the thing workers still have is **a song**.
 - **"Which Side Are You On?"** — **Florence Reece**, Harlan County KY, **1931**. Written on a kitchen calendar the night Sheriff J.H. Blair's men ransacked her house hunting her husband. Set to a Baptist hymn tune. Later picked up by the Freedom Singers in the civil rights movement.
 - **"Sixteen Tons"** — **Merle Travis**, **1946**. The company-store line came from a letter by his brother John. Tennessee Ernie Ford's 1955 cover sold **a million copies in under four weeks**.
 - **"Ludlow Massacre"** — **Woody Guthrie**. A verse-by-verse account of the tent colony and the pit.
 - **"John Henry"** — the steel-driving man who beats the steam drill and dies of it.



WHERE THIS LEAVES US



2025: Starbase, Texas

- On **3 May 2025**, the residents of Boca Chica, Texas voted to incorporate as a city called **Starbase**.
 - Mayor: **Bobby Peden, a SpaceX vice-president**. He ran unopposed. So did both commissioners, both with SpaceX ties
 - SpaceX owns large tracts; ~7,133 more acres slated for annexation



https://en.wikipedia.org/wiki/Starbase,_Texas

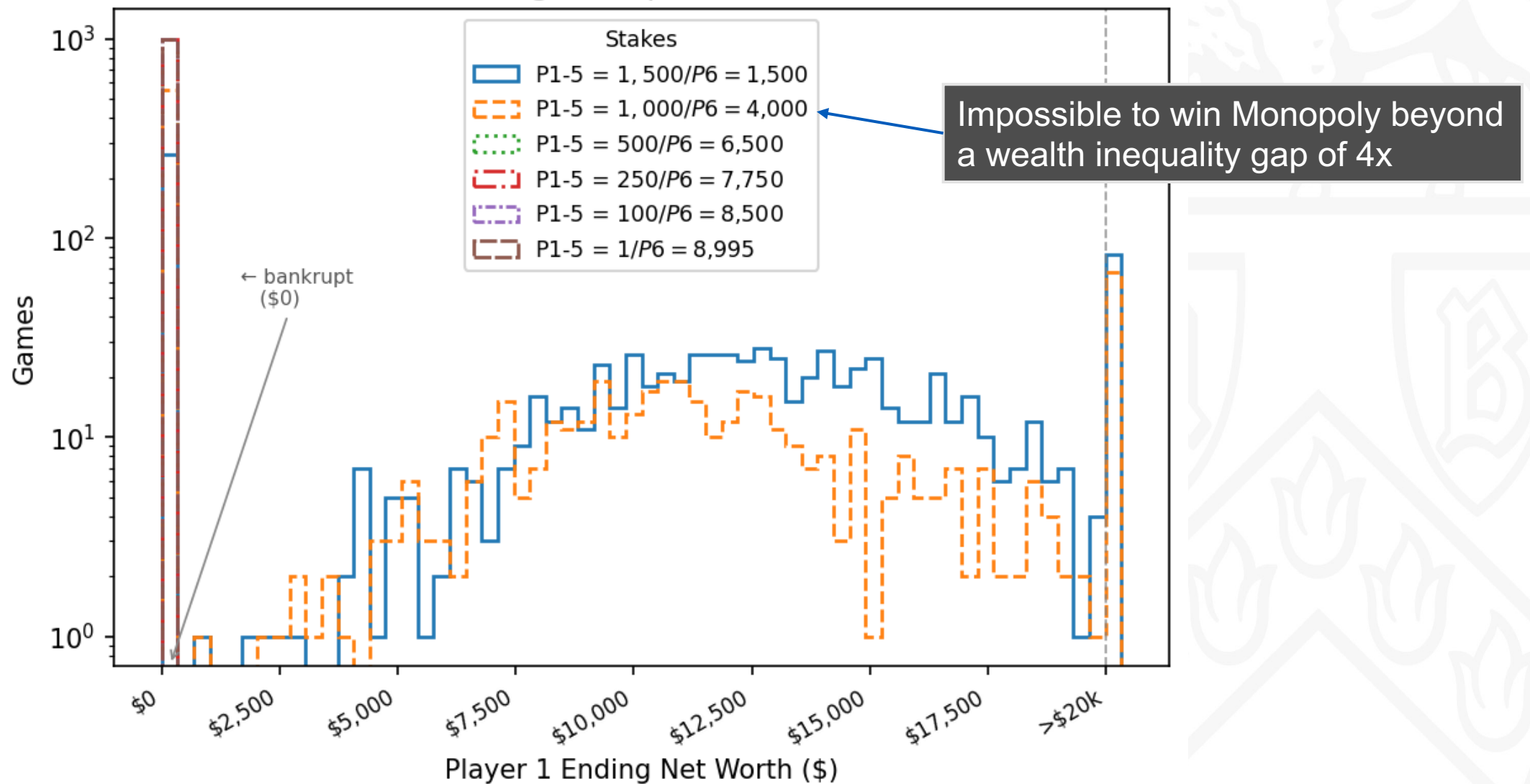
Modern Company Town

The bill always comes due

- **Energy is never just physics.** Every joule we consume has a supply chain, and every supply chain runs through somebody's home.
- **The costs land on whoever has the least power to refuse.** Native Americans, immigrant miners, Black people escaping the Jim Crow south and convict laborers.
- **Losing is a normal outcome.** The miners were right, brave, organized, and defeated by the full backing of the US, State and Local governments as well as privatized paramilitary forces sanctioned by the government.
- **The structures come back.** The company town is being rebuilt right now, in Texas, with better branding.

Data Analysis

Player 1 outcomes as stake advantage shifts to P6
 1000 games per scenario



Sources

- **Wikipedia entry points** (finding aids — trace to the primary sources they cite): Battle of Blair Mountain · Coal Wars · Company town · Coal Region · Susquehannock · Ludlow Massacre · Coal strike of 1902 · Muckraker · Starbase, Texas · Second Industrial Revolution · History of rail transport in the United States · Cornelius Vanderbilt · John D. Rockefeller · J. P. Morgan · Pujo Committee · Lehigh Valley · Theodore Roosevelt · Pan-American Exposition · Rough Riders · California Forever · Ku Klux Klan
- **Institutional / scholarly:** West Virginia Mine Wars Museum · *e-WV: The West Virginia Encyclopedia* (scrip) · Smithsonian Magazine, "What Made the Battle of Blair Mountain the Largest Labor Uprising in American History" · Price V. Fishback, *Journal of Economic History* · Scott Reynolds Nelson, *Steel Drivin' Man* (2006) · Lon Savage, *Thunder in the Mountains* (1990)
- **Contemporary reporting:** Texas Tribune (Starbase incorporation; HB 5246) · Scientific American (Starbase opposition) · New Republic ("Freedom Cities")
- **Watch / read:** *Matewan* (dir. John Sayles, 1987) · Denise Giardina, *Storming Heaven* (1987)